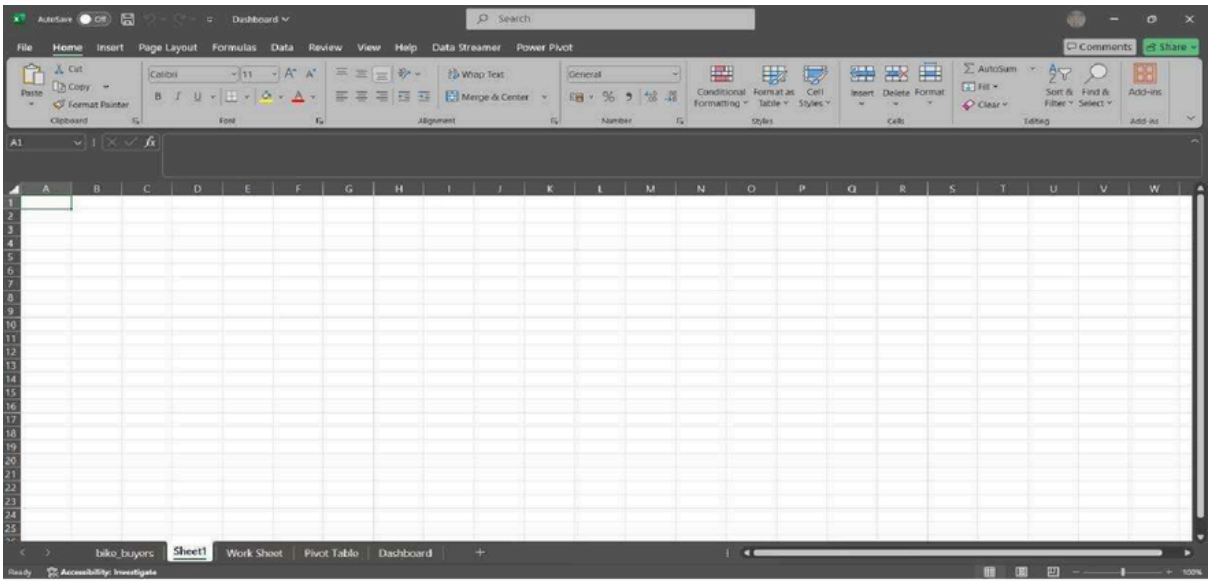
| EX.NO: 12 | DATAANALYSIS AND VISUALIZATION USING ADVANCED EXCEL |
|---|---|
| DATE: | |

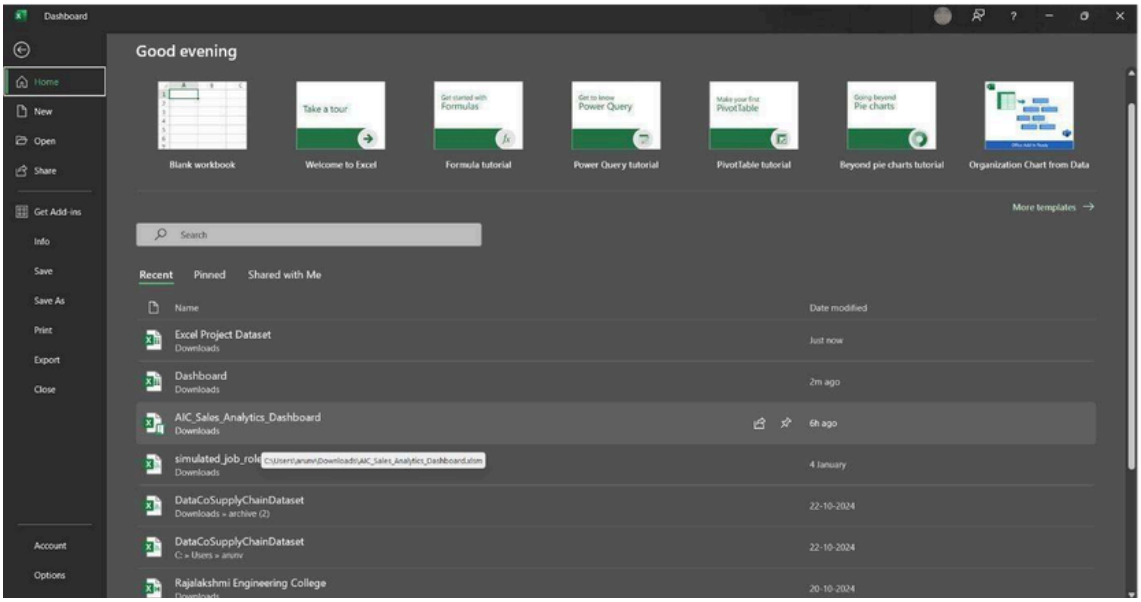
**AIM:**

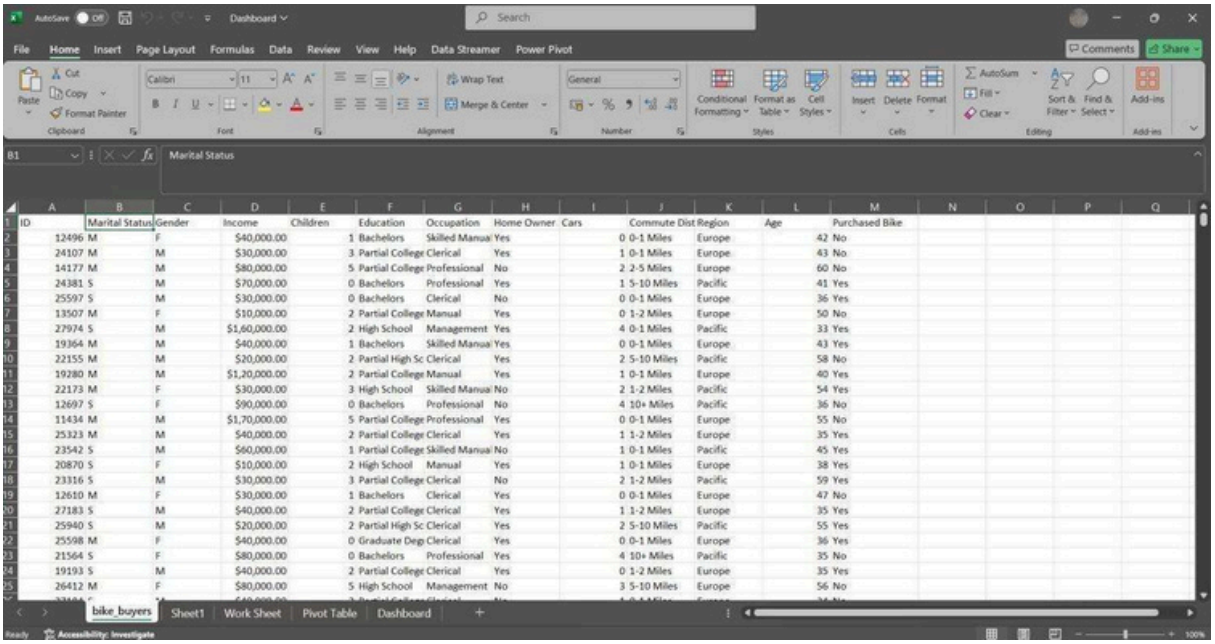
To implement the Data Analysis and Visualization using Advanced Excel.

**PROCEDURE:**

Step 1: Open the blank excel and import the Dataset into Microsoft Excel 2021.

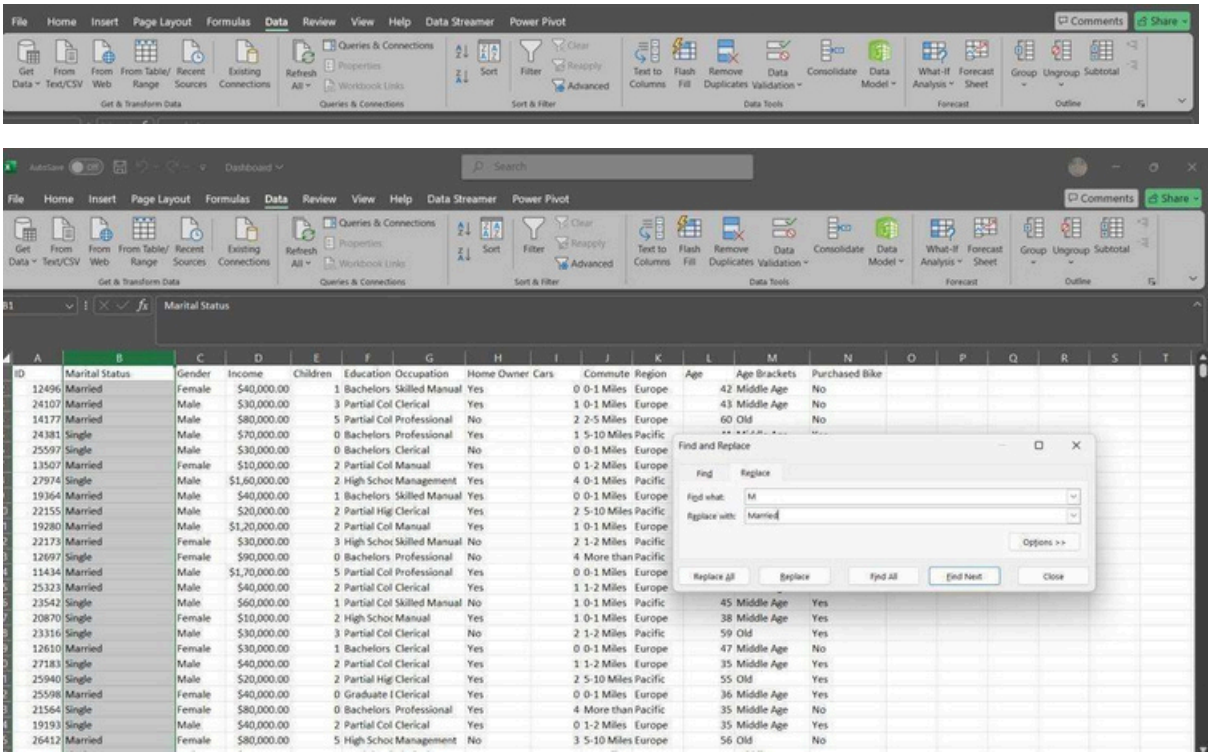




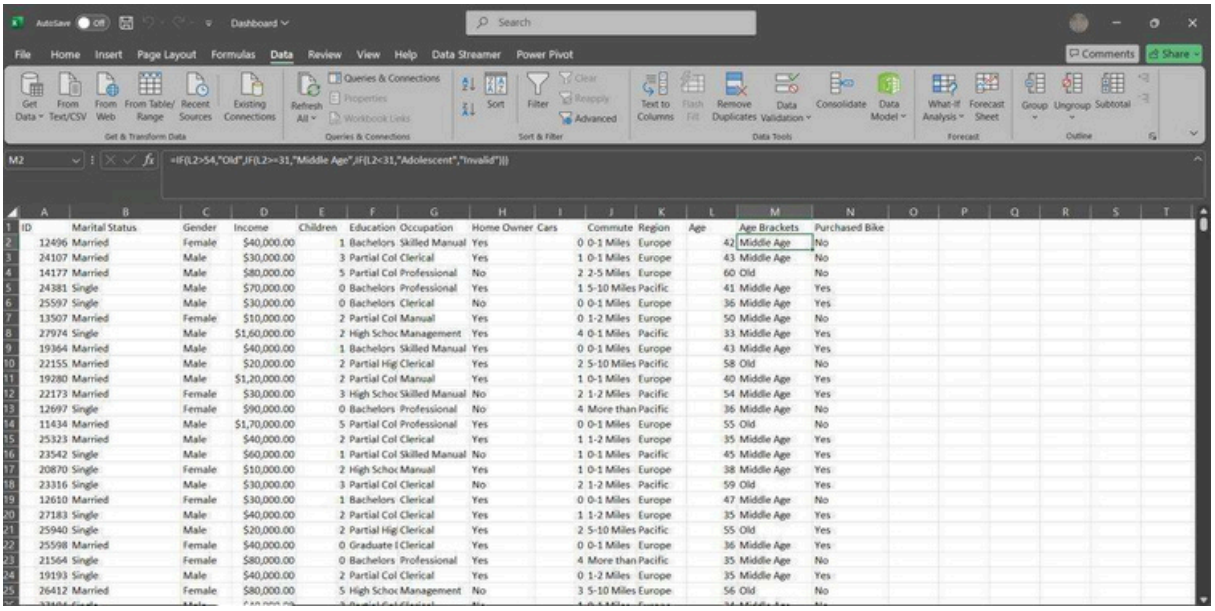
| ID | Marital Status | Gender | Income | Children | Education | Occupation | Home Owner | Cars | Commute Dist | Region | Age | Purchased Bike |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 12496 | M | F | $40,000.00 | 1 | Bachelors | Skilled Manual | Yes | 0 | 0-1 Miles | Europe | 42 | No |
| 24107 | M | M | $30,000.00 | 3 | Partial College | Clerical | Yes | 1 | 0-1 Miles | Europe | 43 | No |
| 14177 | M | M | $80,000.00 | 5 | Partial College | Professional | No | 2 | 2-5 Miles | Europe | 60 | No |
| 24381 | S | M | $70,000.00 | 0 | Bachelors | Professional | Yes | 1 | 5-10 Miles | Pacific | 41 | Yes |
| 25597 | S | M | $30,000.00 | 0 | Bachelors | Clerical | No | 0 | 0-1 Miles | Europe | 36 | Yes |
| 13507 | M | F | $10,000.00 | 2 | Partial College | Manual | Yes | 0 | 1-2 Miles | Europe | 50 | No |
| 27974 | S | M | $1,60,000.00 | 2 | High School | Management | Yes | 4 | 0-1 Miles | Pacific | 33 | Yes |
| 19364 | M | M | $40,000.00 | 1 | Bachelors | Skilled Manual | Yes | 0 | 0-1 Miles | Europe | 43 | Yes |
| 22155 | M | M | $20,000.00 | 2 | Partial High Sc | Clerical | Yes | 2 | 5-10 Miles | Pacific | 58 | No |
| 19280 | M | M | $1,20,000.00 | 2 | Partial College | Manual | Yes | 1 | 0-1 Miles | Europe | 40 | Yes |
| 22173 | M | F | $30,000.00 | 3 | High School | Skilled Manual | No | 2 | 1-2 Miles | Pacific | 54 | Yes |
| 12697 | S | F | $90,000.00 | 0 | Bachelors | Professional | No | 4 | 10+ Miles | Pacific | 36 | No |
| 11434 | M | M | $1,70,000.00 | 5 | Partial College | Professional | Yes | 0 | 0-1 Miles | Europe | 55 | No |
| 25323 | M | M | $40,000.00 | 2 | Partial College | Clerical | Yes | 1 | 1-2 Miles | Europe | 35 | Yes |
| 23542 | S | M | $60,000.00 | 1 | Partial College | Skilled Manual | No | 1 | 0-1 Miles | Pacific | 45 | Yes |
| 20870 | S | F | $10,000.00 | 2 | High School | Manual | Yes | 1 | 0-1 Miles | Europe | 38 | Yes |
| 23316 | S | M | $30,000.00 | 3 | Partial College | Clerical | No | 2 | 1-2 Miles | Pacific | 59 | Yes |
| 12610 | M | F | $30,000.00 | 1 | Bachelors | Clerical | Yes | 0 | 0-1 Miles | Europe | 47 | No |
| 27183 | S | M | $40,000.00 | 2 | Partial College | Clerical | Yes | 1 | 1-2 Miles | Europe | 35 | Yes |
| 25940 | S | M | $20,000.00 | 2 | Partial High Sc | Clerical | Yes | 2 | 5-10 Miles | Pacific | 55 | Yes |
| 25598 | M | F | $40,000.00 | 0 | Graduate Deg | Clerical | Yes | 0 | 0-1 Miles | Europe | 36 | Yes |
| 21564 | S | F | $80,000.00 | 0 | Bachelors | Professional | Yes | 4 | 10+ Miles | Pacific | 35 | No |
| 19193 | S | M | $40,000.00 | 2 | Partial College | Clerical | Yes | 0 | 1-2 Miles | Europe | 35 | Yes |
| 26412 | M | F | $80,000.00 | 5 | High School | Management | No | 3 | 5-10 Miles | Europe | 56 | No |

Step 2: Cleaning and pre-processing the dataset for the better visualization. Using the key Data you can remove the duplicates in the entries. To know clearly about the dataset for the user making changes in row 2 & 3.



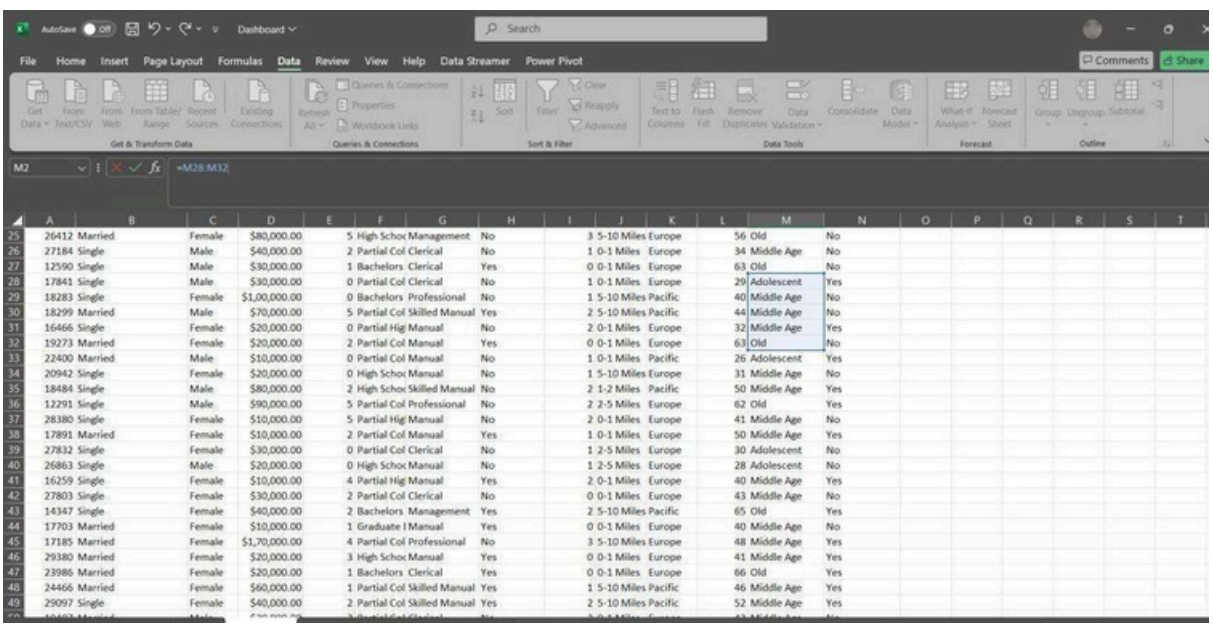
| ID | Marital Status | Gender | Income | Children | Education | Occupation | Home Owner | Cars | Commute | Region | Age | Age Brackets | Purchased Bike |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 12496 | Married | Female | $40,000.00 | 1 | Bachelors | Skilled Manual | Yes | 0 | 0-1 Miles | Europe | 42 | Middle Age | No |
| 24107 | Married | Male | $30,000.00 | 3 | Partial Col | Clerical | Yes | 1 | 0-1 Miles | Europe | 43 | Middle Age | No |
| 14177 | Married | Male | $80,000.00 | 5 | Partial Col | Professional | No | 2 | 2-5 Miles | Europe | 60 | Old | No |
| 24381 | Single | Male | $70,000.00 | 0 | Bachelors | Professional | Yes | 1 | 5-10 Miles | Pacific | | | |
| 25597 | Single | Male | $30,000.00 | 0 | Bachelors | Clerical | No | 0 | 0-1 Miles | Europe | | | |
| 13507 | Married | Female | $10,000.00 | 2 | Partial Col | Manual | Yes | 0 | 1-2 Miles | Europe | | | |
| 27974 | Single | Male | $1,60,000.00 | 2 | High Schoo | Management | Yes | 4 | 0-1 Miles | Pacific | | | |
| 19364 | Married | Male | $40,000.00 | 1 | Bachelors | Skilled Manual | Yes | 0 | 0-1 Miles | Europe | | | |
| 22155 | Married | Male | $20,000.00 | 2 | Partial Hig | Clerical | Yes | 2 | 5-10 Miles | Pacific | | | |
| 19280 | Married | Male | $1,20,000.00 | 2 | Partial Col | Manual | Yes | 1 | 0-1 Miles | Europe | | | |
| 22173 | Married | Female | $30,000.00 | 3 | High Schoo | Skilled Manual | No | 2 | 1-2 Miles | Pacific | | | |
| 12697 | Single | Female | $90,000.00 | 0 | Bachelors | Professional | No | 4 | More than | Pacific | | | |
| 11434 | Married | Male | $1,70,000.00 | 5 | Partial Col | Professional | Yes | 0 | 0-1 Miles | Europe | | | |
| 25323 | Married | Male | $40,000.00 | 2 | Partial Col | Clerical | Yes | 1 | 1-2 Miles | Europe | | | |
| 23542 | Single | Male | $60,000.00 | 1 | Partial Col | Skilled Manual | No | 1 | 0-1 Miles | Pacific | 45 | Middle Age | Yes |
| 20870 | Single | Female | $10,000.00 | 2 | High Schoo | Manual | Yes | 1 | 0-1 Miles | Europe | 38 | Middle Age | Yes |
| 23316 | Single | Male | $30,000.00 | 3 | Partial Col | Clerical | No | 2 | 1-2 Miles | Pacific | 59 | Old | Yes |
| 12610 | Married | Female | $30,000.00 | 1 | Bachelors | Clerical | Yes | 0 | 0-1 Miles | Europe | 47 | Middle Age | No |
| 27183 | Single | Male | $40,000.00 | 2 | Partial Col | Clerical | Yes | 1 | 1-2 Miles | Europe | 35 | Middle Age | Yes |
| 25940 | Single | Male | $20,000.00 | 2 | Partial Hig | Clerical | Yes | 2 | 5-10 Miles | Pacific | 55 | Old | Yes |
| 25598 | Married | Female | $40,000.00 | 0 | Graduate I | Clerical | Yes | 0 | 0-1 Miles | Europe | 36 | Middle Age | Yes |
| 21564 | Single | Female | $80,000.00 | 0 | Bachelors | Professional | Yes | 4 | More than | Pacific | 35 | Middle Age | No |
| 19193 | Single | Male | $40,000.00 | 2 | Partial Col | Clerical | Yes | 0 | 1-2 Miles | Europe | 35 | Middle Age | Yes |
| 26412 | Married | Female | $80,000.00 | 5 | High Schoo | Management | No | 3 | 5-10 Miles | Europe | 56 | Old | No |

Step 3: For the Bike sales we need to capture the specific age group so include a row (Age Brackets) for the better view.



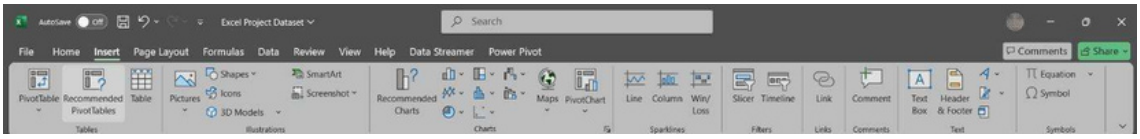
| ID | Marital Status | Gender | Income | Children | Education | Occupation | Home Owner | Cars | Commute | Region | Age | Age Brackets | Purchased Bike |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 12496 | Married | Female | $40,000.00 | 1 | Bachelors | Skilled Manual | Yes | 0 | 0-1 Miles | Europe | 42 | Middle Age | No |
| 24107 | Married | Male | $30,000.00 | 3 | Partial Col | Clerical | Yes | 1 | 0-1 Miles | Europe | 43 | Middle Age | No |
| 14177 | Married | Male | $80,000.00 | 5 | Partial Col | Professional | No | 2 | 2-5 Miles | Europe | 60 | Old | No |
| 24381 | Single | Male | $70,000.00 | 0 | Bachelors | Professional | Yes | 1 | 5-10 Miles | Pacific | 41 | Middle Age | Yes |
| 25597 | Single | Male | $30,000.00 | 0 | Bachelors | Clerical | No | 0 | 0-1 Miles | Europe | 36 | Middle Age | Yes |
| 13507 | Married | Female | $10,000.00 | 2 | Partial Col | Manual | Yes | 0 | 1-2 Miles | Europe | 50 | Middle Age | No |
| 27974 | Single | Male | $1,60,000.00 | 2 | High Schoo | Management | Yes | 4 | 0-1 Miles | Pacific | 33 | Middle Age | Yes |
| 19364 | Married | Male | $40,000.00 | 1 | Bachelors | Skilled Manual | Yes | 0 | 0-1 Miles | Europe | 43 | Middle Age | Yes |
| 22155 | Married | Male | $20,000.00 | 2 | Partial Hig | Clerical | Yes | 2 | 5-10 Miles | Pacific | 58 | Old | No |
| 19280 | Married | Male | $1,20,000.00 | 2 | Partial Col | Manual | Yes | 1 | 0-1 Miles | Europe | 40 | Middle Age | Yes |
| 22173 | Married | Female | $30,000.00 | 3 | High Schoo | Skilled Manual | No | 2 | 1-2 Miles | Pacific | 54 | Middle Age | Yes |
| 12697 | Single | Female | $90,000.00 | 0 | Bachelors | Professional | No | 4 | More than | Pacific | 36 | Middle Age | No |
| 11434 | Married | Male | $1,70,000.00 | 5 | Partial Col | Professional | Yes | 0 | 0-1 Miles | Europe | 55 | Old | No |
| 25323 | Married | Male | $40,000.00 | 2 | Partial Col | Clerical | Yes | 1 | 1-2 Miles | Europe | 35 | Middle Age | Yes |
| 23542 | Single | Male | $60,000.00 | 1 | Partial Col | Skilled Manual | No | 1 | 0-1 Miles | Pacific | 45 | Middle Age | Yes |
| 20870 | Single | Female | $10,000.00 | 2 | High Schoo | Manual | Yes | 1 | 0-1 Miles | Europe | 38 | Middle Age | Yes |
| 23316 | Single | Male | $30,000.00 | 3 | Partial Col | Clerical | No | 2 | 1-2 Miles | Pacific | 59 | Old | Yes |
| 12610 | Married | Female | $30,000.00 | 1 | Bachelors | Clerical | Yes | 0 | 0-1 Miles | Europe | 47 | Middle Age | No |
| 27183 | Single | Male | $40,000.00 | 2 | Partial Col | Clerical | Yes | 1 | 1-2 Miles | Europe | 35 | Middle Age | Yes |
| 25940 | Single | Male | $20,000.00 | 2 | Partial Hig | Clerical | Yes | 2 | 5-10 Miles | Pacific | 55 | Old | Yes |
| 25598 | Married | Female | $40,000.00 | 0 | Graduate I | Clerical | Yes | 0 | 0-1 Miles | Europe | 36 | Middle Age | Yes |
| 21564 | Single | Female | $80,000.00 | 0 | Bachelors | Professional | Yes | 4 | More than | Pacific | 35 | Middle Age | No |
| 19193 | Single | Male | $40,000.00 | 2 | Partial Col | Clerical | Yes | 0 | 1-2 Miles | Europe | 35 | Middle Age | Yes |
| 26412 | Married | Female | $80,000.00 | 5 | High Schoo | Management | No | 3 | 5-10 Miles | Europe | 56 | Old | No |

Step 4: For the different group age peoples we using the formula [=IF(L2>54,"Old",IF(L2>=31,"Middle Age",IF(L2<31,"Adolescent","Invalid")))]

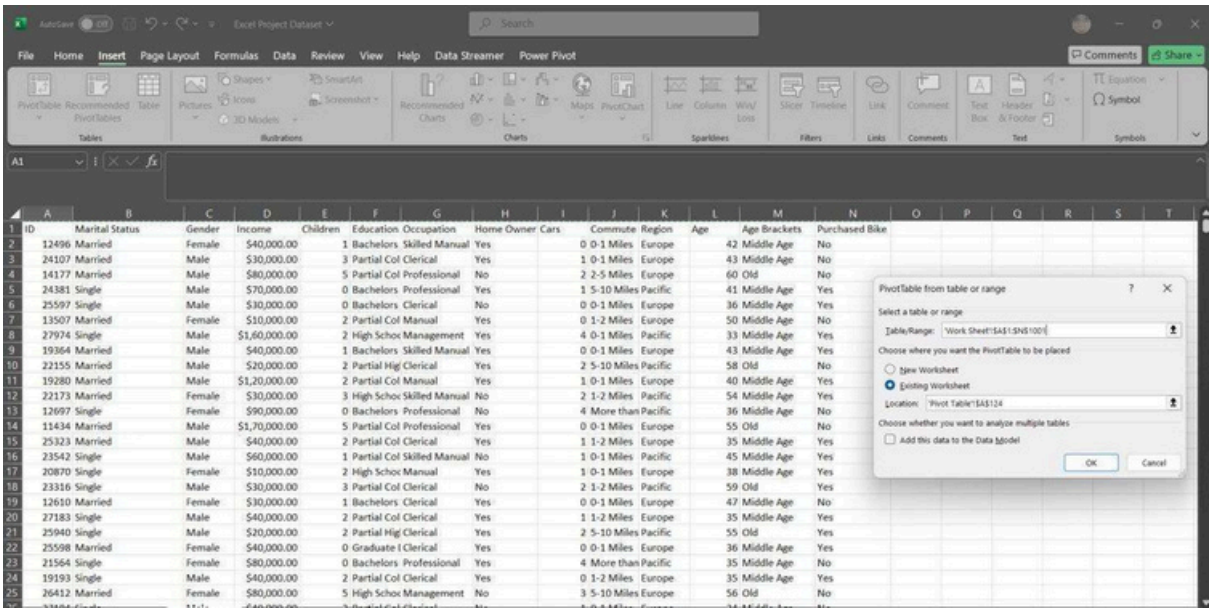


| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 26412 | Married | Female | $80,000.00 | 5 | High Schoo | Management | No | 3 | 5-10 Miles | Europe | 56 | Old | No |
| 27184 | Single | Male | $40,000.00 | 2 | Partial Col | Clerical | No | 1 | 0-1 Miles | Europe | 34 | Middle Age | No |
| 12590 | Single | Male | $30,000.00 | 1 | Bachelors | Clerical | Yes | 0 | 0-1 Miles | Europe | 63 | Old | No |
| 17841 | Single | Male | $30,000.00 | 0 | Partial Col | Clerical | No | 1 | 0-1 Miles | Europe | 29 | Adolescent | Yes |
| 18283 | Single | Female | $1,00,000.00 | 0 | Bachelors | Professional | No | 1 | 5-10 Miles | Pacific | 40 | Middle Age | No |
| 18299 | Married | Male | $70,000.00 | 5 | Partial Col | Skilled Manual | Yes | 2 | 5-10 Miles | Pacific | 44 | Middle Age | No |
| 16466 | Single | Female | $20,000.00 | 0 | Partial Hig | Manual | No | 2 | 0-1 Miles | Europe | 32 | Middle Age | Yes |
| 19273 | Married | Female | $20,000.00 | 2 | Partial Col | Manual | Yes | 0 | 0-1 Miles | Europe | 63 | Old | No |
| 22400 | Married | Male | $10,000.00 | 0 | Partial Col | Manual | No | 1 | 0-1 Miles | Pacific | 26 | Adolescent | Yes |
| 20942 | Single | Female | $20,000.00 | 0 | High Schoo | Manual | No | 1 | 5-10 Miles | Europe | 31 | Middle Age | No |
| 18484 | Single | Male | $80,000.00 | 2 | High Schoo | Skilled Manual | No | 2 | 1-2 Miles | Pacific | 50 | Middle Age | Yes |
| 12291 | Single | Male | $90,000.00 | 5 | Partial Col | Professional | No | 2 | 2-5 Miles | Europe | 62 | Old | Yes |
| 28380 | Single | Female | $10,000.00 | 5 | Partial Hig | Manual | No | 2 | 0-1 Miles | Europe | 41 | Middle Age | No |
| 17891 | Married | Female | $10,000.00 | 2 | Partial Col | Manual | Yes | 1 | 0-1 Miles | Europe | 50 | Middle Age | Yes |
| 27832 | Single | Female | $30,000.00 | 0 | Partial Col | Clerical | No | 1 | 2-5 Miles | Europe | 30 | Adolescent | No |
| 26863 | Single | Male | $20,000.00 | 0 | High Schoo | Manual | No | 1 | 2-5 Miles | Europe | 28 | Adolescent | No |
| 16259 | Single | Female | $10,000.00 | 4 | Partial Hig | Manual | Yes | 2 | 0-1 Miles | Europe | 40 | Middle Age | Yes |
| 27803 | Single | Female | $30,000.00 | 2 | Partial Col | Clerical | No | 0 | 0-1 Miles | Europe | 43 | Middle Age | No |
| 14347 | Single | Female | $40,000.00 | 2 | Bachelors | Management | Yes | 2 | 5-10 Miles | Pacific | 65 | Old | Yes |
| 17703 | Married | Female | $10,000.00 | 1 | Graduate I | Manual | Yes | 0 | 0-1 Miles | Europe | 40 | Middle Age | No |
| 17185 | Married | Female | $1,70,000.00 | 4 | Partial Col | Professional | No | 3 | 5-10 Miles | Europe | 48 | Middle Age | Yes |
| 29380 | Married | Female | $20,000.00 | 3 | High Schoo | Manual | Yes | 0 | 0-1 Miles | Europe | 41 | Middle Age | Yes |
| 23986 | Married | Female | $20,000.00 | 1 | Bachelors | Clerical | Yes | 0 | 0-1 Miles | Europe | 66 | Old | Yes |
| 24466 | Married | Female | $60,000.00 | 1 | Partial Col | Skilled Manual | Yes | 1 | 5-10 Miles | Pacific | 46 | Middle Age | Yes |
| 29097 | Single | Female | $40,000.00 | 2 | Partial Col | Skilled Manual | Yes | 2 | 5-10 Miles | Pacific | 52 | Middle Age | Yes |

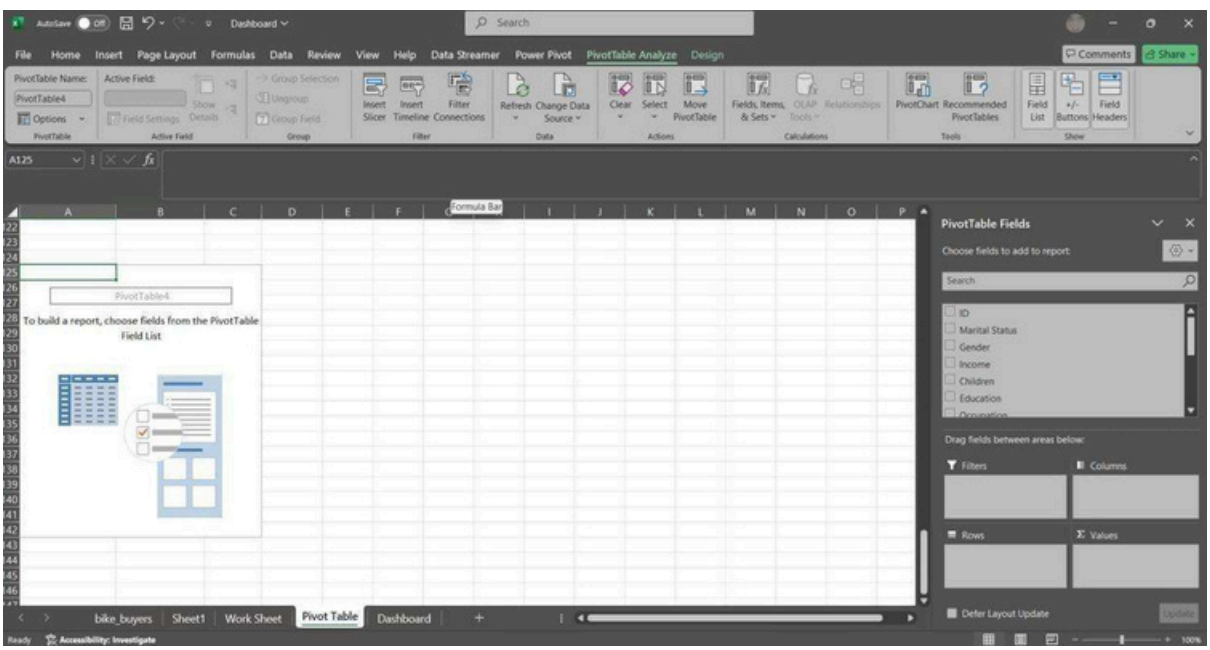
Step 5: After the completion of the data cleaning and pre-processing now we can start to create the pivot table. Make an add of a sheet in the same board to work pivot table in different table.



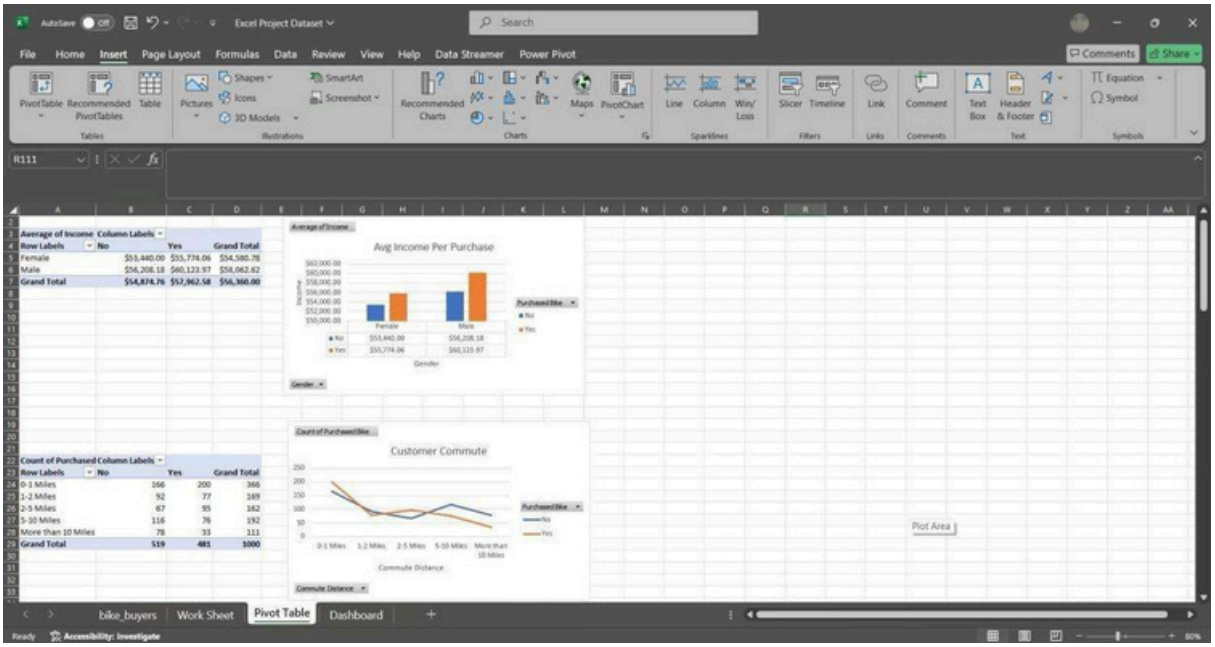
Step 6: Select the pivot table and in the down, you can see the dataset select that sheet and give ctrl+a to work on the pivot table.



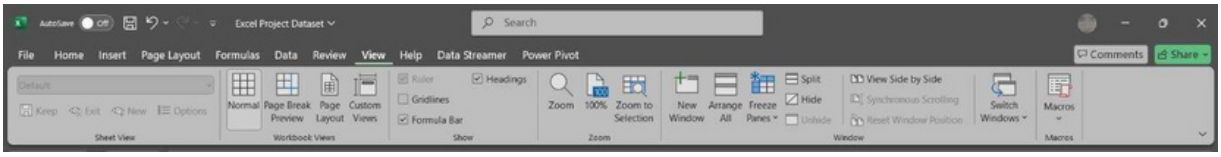
Step 7: For the user needs we can select and drag the option to make chart with following datasets to make understand the user we can use charts too.



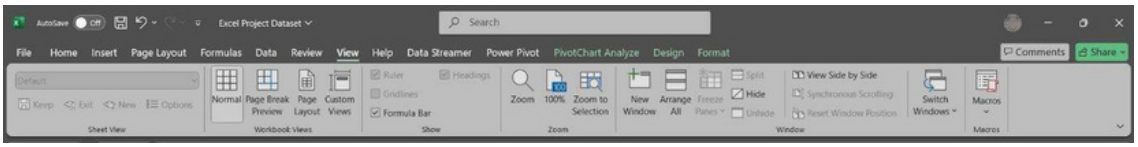
Step 8: Using recommended charts we can visualize the total value which we taken in pivot table.



Step 9: Now creating a Dashboard in the excel sheet same like pivot table add a sheet for dashboard. Using the view option you can edit the grid lines.



Step 10: Design the dashboard to look with informative and user friendly. Using pivot chart analysis option, you can work on the design and slicer to know about the detail information of pivot chart.



**RESULT:**

Thus the Data Analysis and Visualization using Advanced Excel was implemented successfully.